

Skin Manifestations of Onchocerciasis

Skin involvement in onchocerciasis can present in various forms, including **acute papular onchodermatitis**, **chronic papular onchodermatitis**, **lichenified onchodermatitis**, **atrophy**, and **depigmentation**. Palpable onchocercal nodules may also be present. More than one pattern of skin involvement may occur simultaneously, or one pattern may evolve into another over time.

Pruritus is often the first symptom of infection.

Acute Papular Onchodermatitis

- Typically affects the **face**, **extremities**, and **trunk**
- Characterized by widespread **small pruritic papules**, **vesicles**, and **pustules**
- May be accompanied by **erythema** and **edema**
- A similar rash can be precipitated by treatment with **ivermectin** or **diethylcarbamazine (DEC)** in infected individuals

Chronic Papular Onchodermatitis

- Lesions consist of **macules** and **lichenoid papules**, 3–9 mm in diameter
- **Pruritus** is common
- **Post-inflammatory hyperpigmentation** may develop
- Most frequently involves the **buttocks**, **shoulders**, and **waist**

Other Skin Changes

- **Lichenified onchodermatitis**: results from chronic scratching and rubbing
- **Atrophy**: resembles age-related skin atrophy

- **Depigmentation:** may mimic post-inflammatory hypopigmentation or vitiligo

Related Physical Findings

Palpable onchocercal nodules: - Contain adult worms within the **deep dermis and subcutaneous tissue** - Typically **asymptomatic** - Located over bony prominences such as the **skull, iliac crest, knee, ribs, sacrum, scapula, and trochanter**

Laboratory Tests

- **Peripheral eosinophilia** is common
- **Elevated serum IgE levels** are frequently observed

Special Tests

- **Diagnosis is most commonly confirmed by:** - Identification of **microfilariae** in a **skin snip** - Visualization of an **adult worm** in an excised nodule
- **Slit-lamp examination** of the eye may reveal microfilariae in the **anterior chamber**
- **Antibody testing** lacks specificity
- **PCR testing** is available primarily through research laboratories

Differential Diagnosis

The differential diagnosis varies based on clinical presentation:

- **Pruritus alone:** consider **scabies**
- **Acute papular onchodermatitis:** may resemble **miliaria, insect bites, or eczema**
- **Chronic papular onchodermatitis:** can mimic **chronic eczema, lichenoid dermatitis, or atopic dermatitis**

- **Lichenified onchodermatitis:** differential includes any condition causing chronic pruritus or friction (e.g., lichen simplex chronicus)
- **Skin atrophy:** consider normal aging or other causes of dermal atrophy
- **Depigmentation:** differentiate from post-inflammatory hypopigmentation or **vitiligo**
- **Onchocercal nodules:** consider other parasitic nodules (e.g., cysticercosis) or **epidermal inclusion cysts**

Treatment

Ivermectin is the drug of choice for onchocerciasis.

- **Diethylcarbamazine (DEC)** should **not** be used in confirmed cases due to:
 - Lack of efficacy against adult worms
 - Risk of severe **ocular reactions**
- **Mazzotti test** (diagnostic use of DEC):
 - A single **50-mg dose of DEC** may be used in areas where **loiasis is not endemic**
 - In infected individuals, it provokes a reaction within hours:
 - Worsening of cutaneous symptoms
 - Migration of microfilariae to the dermis
 - Allows confirmation via skin biopsy

Ivermectin effectively reduces microfilarial load and prevents disease progression, though it does not reliably kill adult worms.